

Shenglai Zeng

🌐 Personal Website

🎓 Google Scholar

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RESEARCH INTEREST

I am a third-year PhD student at Michigan State University advised by Professor Jiliang Tang. My current research interests are mainly about Trustworthy AI, large language models(LLMs) and Information retrieval (IR). I also worked on federated learning for years. I have won the Best Paper Award of IEEE Transactions on Cloud Computing, 2023.

EDUCATION

Michigan State University

DSE Lab/PhD students in Computer Science and Engineering

East Lansing, U.S

Sept 2023-Present

Advisor: Jiliang Tang

Lab: Data Science and Engineering Lab

Research Direction: Trustworthy AI, Large language models(LLMs)

University of Electronic Science and Technology of China

Yingcai Honor School/B.Sc in Computer Science and Engineering

Chengdu, China

Sept 2019-Present

CGPA: 3.98/4.00

Weighted Average: 93.97/100(1st among 100 students)

Honors: The Most Outstanding Students Award of UESTC

PREPRINTS

- **Shenglai Zeng**, Tianqi Zheng, Chuan Tian, Dante Everaert, Yau-Shian Wang, Yupin Huang, Michael J. Morais, Rohit Patki, Jinjin Tian, Xinnan Dai, Kai Guo, Monica Xiao Cheng, Hui Liu
Attn-GS: Attention-Guided Context Compression for Efficient Personalized LLMs
Submitted to ACL ARR
- Kai Guo, Xinnan Dai, **Shenglai Zeng**, Harry Shomer, Haoyu Han, Yu Wang, Jiliang Tang
Beyond Static Retrieval: Opportunities and Pitfalls of Iterative Retrieval in GraphRAG
Submitted to ICLR
- Xinnan Dai, Kai Guo, Chung-Hsiang Lo, **Shenglai Zeng**, Jiayuan Ding, Dongsheng Luo, Subhabrata Mukherjee, Jiliang Tang
GraphGhosts: Tracing Reasoning Structures Behind Large Language Models
Submitted to ACL ARR
- Yuping Lin, Pengfei He, Haoran Zhao, **Shenglai Zeng**, Zhen Xiang, Hui Liu, Jiliang Tang
Structural Alignment Faking: Hiding Malicious Capabilities in Mixture-of-Experts LLMs
Pre-print
- Jie Ren, Han Xu, Pengfei He, Yingqian Cui, **Shenglai Zeng**, Jiankun Zhang, Hongzhi Wen, Jiayuan Ding, Hui Liu, Yi Chang, Jiliang Tang
Copyright Protection in Generative AI: A Technical Perspective
Pre-print

PUBLICATIONS

- Jie Ren, Zhenwei Dai, Xianfeng Tang, Yue Xing, **Shenglai Zeng**, Hui Liu, Jingying Zeng, Qiankun Peng, Samarth Varshney, Suhang Wang, Qi He, Charu C. Aggarwal, Hui Liu

Keeping an Eye on LLM Unlearning: The Hidden Risk and Remedy
NIPS-2025

- Shenglai Zeng, Jiankun Zhang, Pengfei He, Jie Ren, Tianqi Zheng, Hanqing Lu, Han Xu, Hui Liu, Yue Xing, Jiliang Tang
Mitigating the privacy issues in retrieval-augmented generation (rag) via pure synthetic data
EMNLP-2025-main
- Jiankun Zhang*, Shenglai Zeng*, Jie Ren, Tianqi Zheng, Hui Liu, Xianfeng Tang, Yi Chang
Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation
EMNLP-2025-main
- Kai Guo, Harry Shomer, Shenglai Zeng, Haoyu Han, Yu Wang, Jiliang Tang
Empowering GraphRAG with Knowledge Filtering and Integration
EMNLP-2025-main
- Shenglai Zeng, Pengfei He, Kai Guo, Tianqi Zheng, Hanqing Lu, Yue Xing, Hui Liu
Towards Context-Robust LLMs: A Gated Representation Fine-tuning Approach
ACL-2025-main
- Bo Wang, Weiye He, Pengfei He, Shenglai Zeng, Zhen Xiang, Yue Xing, Jiliang Tang
Unveiling privacy risks in llm agent memory
ACL-2025-main
- Bingheng Li, Zhikai Chen, Haoyu Han, Shenglai Zeng, Jingzhe Liu, Jiliang Tang
Unveiling Mode Connectivity in Graph Neural Networks
KDD-2025
- Shenglai Zeng, Jiankun Zhang, Bingheng Li, Yuping Lin, Tianqi Zheng, Dante Everaert, Hanqing Lu, Hui Liu, Yue Xing, Monica Xiao Cheng, Jiliang Tang
Towards Knowledge Checking in Retrieval-augmented Generation: A Representation Perspective
NAACL-2025-main(Oral)
- Jie Ren, Kangrui Chen, Yingqian Cui, Shenglai Zeng, Hui Liu, Yue Xing, Jiliang Tang, Lingjuan Lyu
Six-cd: Benchmarking concept removals for benign text-to-image diffusion models
CVPR-2025
- Pengfei He, Yue Xing, Han Xu, Jie Ren, Yingqian Cui, Shenglai Zeng, Jiliang Tang, Makoto Yamada, Mohammad Sabokrou
Stealthy Backdoor Attack via Confidence-driven Sampling
TMLR
- Jie Ren, Yaxin Li, Shenglai Zeng, Han Xu, Lingjuan Lyu, Yue Xing, Jiliang Tang
Unveiling and mitigating memorization in text-to-image diffusion models through cross attention
ECCV-2024
- Han Xu, Jie Ren, Pengfei He, Yingqian Cui, Shenglai Zeng, Hui Liu, Jiliang Tang, Amy Liu
On the Generalization of Training-based ChatGPT Detection Methods
EMNLP-2024-Findings
- Shenglai Zeng*, Yaxin Li*, Jie Ren, Yiding Liu, Han Xu, Pengfei He, Yue Xing, Shuaiqiang Wang, Jiliang Tang, Dawei Yin
Exploring Memorization in Fine-tuned Language Models
ACL-2024
- Shenglai Zeng*, Jiankun Zhang*, Pengfei He, Yue Xing, Yiding Liu, Han Xu, Jie Ren, Shuaiqiang Wang, Dawei Yin, Yi Chang, Jiliang Tang
The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)
ACL-2024-findings
- Pengfei He, Han Xu, Jie Ren, Yingqian Cui, Shenglai Zeng, Hui Liu, Charu Aggarwal, Jiliang Tang
Sharpness-aware Data Poisoning Attack

ICLR-2024 Spotlight

- Juanhui Li, Harry Shomer, Haitao Mao, **Shenglai Zeng**, Yao Ma, Neil Shah, Jiliang Tang, Dawei Yin
Evaluating graph neural networks for link prediction: Current pitfalls and new benchmarking
NIPS-2023 Benchmark
- **Shenglai Zeng**, Zonghang Li, Hongfang Yu, Zhihao Zhang, Long Luo, Bo Li, Dusit Niyato
HFedMS: Heterogeneous Federated Learning with Memorable Data Semantics in Industrial Meta-verse
IEEE Transactions on Cloud Computing, 2023 Best Paper
- **Shenglai Zeng**, Zonghang Li, Hongfang Yu, Yihong He, Zenglin Xu, Dusit Niyato, Han Yu
Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training
International Conference on Database Systems for Advanced Applications (DASFAA-2022)
- Jiaqi Wang*, **Shenglai Zeng***, Zewei Long, Yaqing Wang, Houping Xiao, Fenglong Ma
Knowledge-Enhanced Semi-Supervised Federated Learning for Aggregating Heterogeneous Lightweight Clients in IoT
SDM-2023

Patent

- **Shenglai Zeng**, Zonghang Li, Yihong He, Xun Zhang, Hongfang Yu, Gang Sun
"A Hierarchical User Training Management Architecture and Training Strategy for Non-i.i.d Data" Chinese patent

RESEARCH EXPERIENCE

DSE Lab, Michigan State University

Research Assistant / Research on Trustworthy AI and LLM-safety

Lansing, U.S

Sept 2023 - Present

-Advisor: Professor Jiliang Tang

- Identify and mitigate the real privacy issues of LLMs.
- Diverse attack/defense techniques on LLM systems.
- Deeper understanding of underlying mechanism behind LLMs, especially knowledge extraction perspective.
- Leverage LLMs to enhance/empower challenging applications and tasks.

Search Science Team, Amazon

Applied Scientist Intern / Research on Retrieval-augmented Generation and personalization

CA, USA

Summer 2024, 2025

-Mentor: Tianqi Zheng, Dante Everaert

-Manager: Hanqing Lu, Chuan Tian, Monica Xiao Cheng

- Representation finetuning for robust RAG [ACL-2025].
- Utilize LLMs' internal behavior to conduct knowledge checking in RAG [NAACL-2025(oral)].
- Privacy of multi-modality RAG [EMNLP-2025]
- Synthetic data for RAG [EMNLP-2025]

Search Science Team, Baidu.Inc

Research Intern / Research on the memorization of LLM

Beijing, China

May 2023 - May 2024

-Mentor: Dr. Yiding Liu and Dr. Dawei Yin

- Investigating the memorization behavior and privacy implications of fine-tuned LLMs [ACL-2024].
- Currently worked on privacy risks of Retrieval LMs and AI-agents [ACL-2024].

Intelligent Networking and Applications Research Center, UESTC

Research Assistant / Research on the Optimization of Federated Learning

Chengdu, China

Sept 2020 - Jun 2023

-Mentor: Professor Hongfang Yu and Dr. Zonghang Li

- Proposed a novel idea of Sequential-to-Parallel training in FL.

- One conference Paper accepted by DASFAA 2022(First author)
- Investigated the application of FL in Industrial Metaverse.
- One journal paper accepted by IEEE TCC(First author).

The Pennsylvania State University

Pennsylvania, USA

Online Intern/Research on Semi-supervised Federated Learning

Jun 2021 - Jun 2022

-Mentor: Professor Fenglong Ma

- Implemented a semi-supervised federated learning system combined with novel personalized punning and structure-aware collaborative distillation techniques.
- Paper accepted by to SDM 2022(Co-First Author)
- Currently focusing on FL with different model structures.

University of British Columbia

Vancouver, Canada

Summer Intern/Federated Data Evaluation with Unlearning

Jun 2022 - Sept 2022

-Mentor: Professor Xiaoxiao Li

- Try to use the concept of cooperative game to evaluate the importance of data of participating users in federated learning.
- Try to accelerate the evaluation process and straggler problem using federated unlearning.

The University of Chicago

Chicago, USA

Online Intern/Research on the IOT & Sensing Security

Mar 2020 - Mar 2021

-Mentor: Shinan Liu(PhD candidate)

- Tried to use audio data collected by microphone to reconstruct user's state of motion during recording time.
- Proposed a mathematical-physical model to explain the correlation between different sensors' responses to motion.

AWARDS AND ACHIEVEMENTS

- Best Paper Award, IEEE Transactions on Cloud Computing (JCR Q1)
- Spotlight Paper Award, ICLR
- The Most Outstanding Students Award of UESTC (Highest honor in UESTC, Only 10 students are awarded)
- National Scholarship in the session of 2019-2020.(Highest honor of undergraduate student)
- WAC Scholarship in the session of 2020-2021.(Only 10 undergraduate students in UESTC are awarded each year)
- 1st Outstanding Academic Scholarship in 2020,2021,and 2022.(Top 5 % students)

Services

- Program Committee or Reviewer:** IEEE TMC, IEEE Trans on Service Computing, IEEE TPDS, IEEE TKDE, IoT-J, IEEE Trans on Information Forensics Security, MICCAI Workshop on Distributed, Collaborative and Federated Learning (DeCaF-2022), The ACM Transactions on Information Systems (TOIS), IEEE Transactions on Vehicular Technology (TVT), IEEE TKDD, ACL-ARR, NIPS
- Reviews 30+ papers.